

University of Florence
Department of Information Engineering (DINFO)

HYDRAS Project Weekly Report

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The value of the formation: single agent vs. team

Retraining ablation and comparative explainability

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Abstract. To quantify the importance of the master–slave *formation* we trained a *single* agent, without formation, keeping everything else equal: starting from the single-ring model ($v_{\max} = 2$ m/s) we fine-tune it (2 M steps) with the directional sensors **zeroed** (**mask_formation**). The agent keeps only the concentration at its own position, the concentration/displacement memories, wind and current. The success-rate drop measures the *net* value of the formation (unlike a frozen-model ablation, here the agent can adapt). We then run the full explainability protocol on *both* models. Result: the formation is worth ~ 6 SR points overall, concentrated in the diffuse-plume scenarios, but the retrained single agent stays far above the gradient method (FCM); the explainability shows the agent shifting from *spatial* to *temporal* sensing to compensate.

1 Inference results of the single agent

The no-formation model is evaluated with the standard held-out protocol: 26 unseen sources (SRC107–SRC132), 4 wind scenarios (V0–V3), 3 temporal chunks (Q1/4, Q1/2, Q3/4), 5 episodes per configuration — 1560 episodes. Table 1 reports the success rate by scenario and chunk.

Metric	Value	Episodes
Global SR	93.3%	1560
Mean steps to success	100.6	—
Wind V0 (calm)	97.2%	390
Wind V1 (NW)	97.9%	390
Wind V2 (SW)	86.4%	390
Wind V3 (E)	91.8%	390
Chunk Q1/4 (compact plume)	97.9%	520
Chunk Q1/2 (intermediate plume)	90.0%	520
Chunk Q3/4 (dispersed plume)	92.1%	520

Table 1: Held-out inference of the single agent (no formation, $v_{\max}=2$ m/s).

Failures concentrate where the plume is most dispersed and the current strongest: **V2** (86.4%) and the intermediate chunk **Q1/2** (90.0%) — the same critical scenarios for every approach. Figure 1 shows the detail by wind–chunk combination.

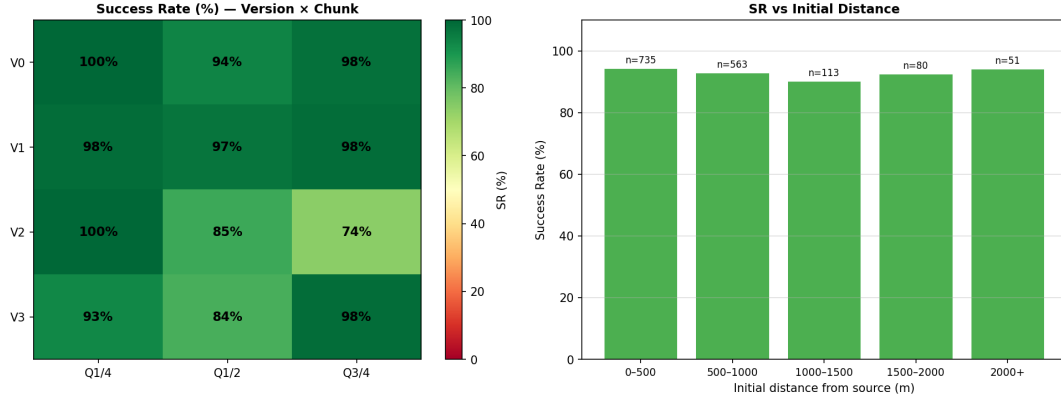


Figure 1: Success rate broken down by wind scenario (V0–V3) and temporal chunk (Q1/4, Q1/2, Q3/4) — single agent (no formation). Each bar is the SR over the held-out sources for that condition; the residual difficulties concentrate on the advected/diffuse regimes V2 and Q1/2.

2 Comparison with the model *with* formation

The comparison is at parity of everything (same speed $v_{\max}=2$, same architecture, same protocol): the only difference is the presence or absence of the formation. The model with formation is the single-ring $v_{\max}=2$ model from which the no-formation one was fine-tuned.

Metric	With formation	Without formation	Δ
Global SR	99.7%	93.3%	−6.4
Mean steps (succ.)	69.5	100.6	+44.7%
V0	100.0%	97.2%	−2.8
V1	99.5%	97.9%	−1.6
V2	99.2%	86.4%	−12.8
V3	100.0%	91.8%	−8.2
Q1/4	100.0%	97.9%	−2.1
Q1/2	99.0%	90.0%	−9.0
Q3/4	100.0%	92.1%	−7.9

Table 2: With vs. without formation ($v_{\max}=2$ m/s, 1560 episodes each). Δ is the cost of removing the formation.

The formation is worth **~6.4 points** of global success rate and about 45% in speed. The cost is **not uniform**: it is negligible in the easy scenarios (V0/V1, Q1/4, within −3 points) and becomes marked where the instantaneous *spatial* gradient matters most — **V2** (−12.8) and **Q1/2** (−9.0), the advected/diffuse-plume regimes. It is precisely there that the eight simultaneous directional readings of the formation let the agent

estimate the source direction in one shot, whereas the single agent must reconstruct it over time. The drop, however, is **moderate**: the retrained single agent recovers most of the performance by adapting its strategy (Section 4).

3 Comparison with the gradient method (FCM)

The most instructive comparison is against the adaptive FCM at equal maximum speed ($lr = 20 \text{ m} \approx v_{\max}=2 \text{ m/s}$). The FCM *uses the formation* (the 9 sensors to estimate the gradient), the single agent *does not*: yet Table 3 shows that the single agent, though deprived of the formation, clearly outperforms the FCM that has it.

SR	FCM (with form.)	RL no form.	RL with form.
Global	58.2%	93.3%	99.7%
V0	80.1%	97.2%	100.0%
V1	32.1%	97.9%	99.5%
V2	34.6%	86.4%	99.2%
V3	85.9%	91.8%	100.0%
Q1/2	38.0%	90.0%	99.0%

Table 3: Single agent (RL without formation) vs. adaptive FCM with formation ($lr=20$) vs. RL with formation, at equal speed ($v_{\max}=2$).

The single agent beats the FCM by +35 **points** of global success rate, and the gap explodes in the adverse scenarios (V1: +66, V2: +52, Q1/2: +52). The message is strong: **the learned strategy is worth more than the formation itself**. An RL agent that sees only its own concentration, but knows how to integrate it over time and remember where the signal was strongest, localizes the source far better than a reactive method that has the full sensor ring. The formation helps (the +6.4 points up to the full model), but it is not what separates success from failure: that is *learning*.

4 Explainability of the single agent

What does the agent rely on *without* the formation? We apply the full explainability protocol: causal group ablation (624 paired spawns; a feature group is zeroed and SR/steps are re-measured, with 95% bootstrap confidence intervals over the paired spawns), exact Shapley values (2^{10} coalitions over 1500 states), and behavioural probes.

4.1 Causal group ablation

Ablated group	Δ SR	95% CI	Δ steps
Displacements (history)	−55.6	[−59.8, −51.8]	+320%
Vector to max conc.	−25.8	[−29.3, −22.1]	+4%
Current (u, v)	−12.8	[−16.5, −9.5]	+40%
Concentration (history)	−11.2	[−14.3, −8.3]	+90%
Max conc. value	−7.1	[−9.8, −4.5]	+31%
Concentration (instant)	−3.2	[−5.6, −0.8]	+54%
Wind (u, v)	−1.4	[−3.7, +0.6]	+3%
Age since last contact	−0.5	[−2.4, +1.3]	−6%

Table 4: Causal group ablation on the single agent (baseline SR 93.6%); genuine, non-masked feature groups only.

Without the formation the agent relies overwhelmingly on the **displacement memory**: zeroing it costs **−55.6 points** of SR and over 320% in steps — by far the most load-bearing group. Next come the **vector to the maximum concentration seen** (−25.8, i.e. memory-based *homing*), the **current** (−12.8) and the **history of its own concentration** (−11.2). Deprived of the formation’s instantaneous *spatial* gradient, the agent reconstructs it in *time*, correlating its own displacements with the concentration changes it measures beneath itself, and heads towards the “hottest” point encountered so far.

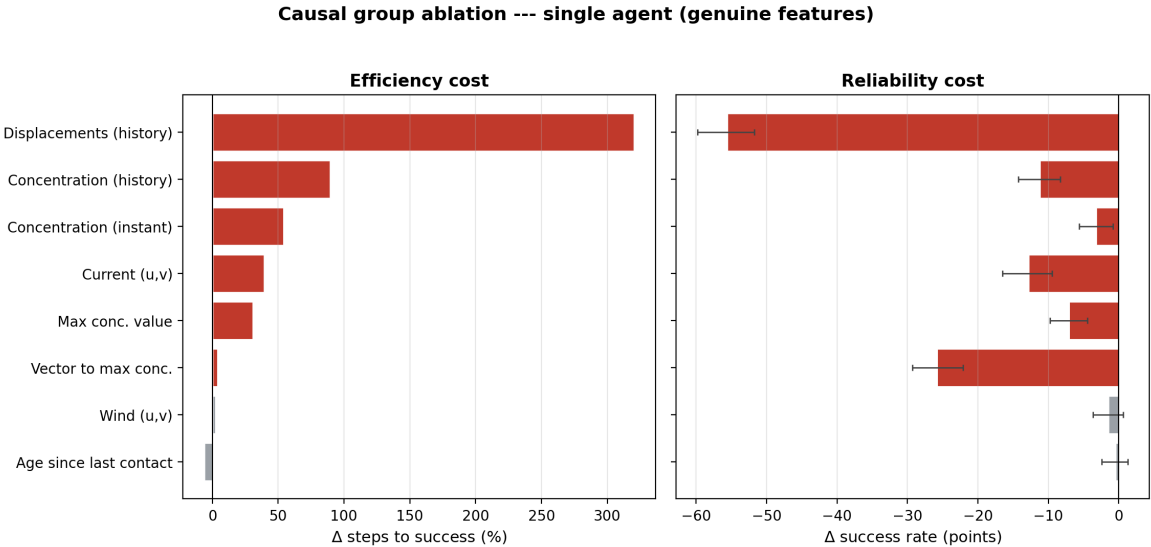


Figure 2: Group ablation — single agent (genuine, non-masked groups). Left: efficiency cost (steps); right: reliability cost (SR); red when the interval excludes zero.

Note on the ring artifact. The ring groups are **excluded** from both Table 4 and Figures 2–3. The ring sensors are zeroed (raw = 0) throughout the single agent’s training and inference, so they carry no information. However, `VecNormalize` normalizes them against the mean/variance inherited from the model with formation,

turning the zero into a non-zero *constant* (~ 0.41 in normalized space); ablating that constant would perturb the learned *bias*, yielding a spurious ΔSR ($-13/-16$) that reflects sensitivity to a constant, not information content — hence their exclusion.

4.2 Exact Shapley values

The Shapley values confirm the ablation ranking: the displacement memory dominates with $|\phi| = 0.27$, followed by current (0.12) and concentration history (0.08); the ten groups jointly account for $0.50 - 0.17 = 0.33$ of probability mass on the chosen action.

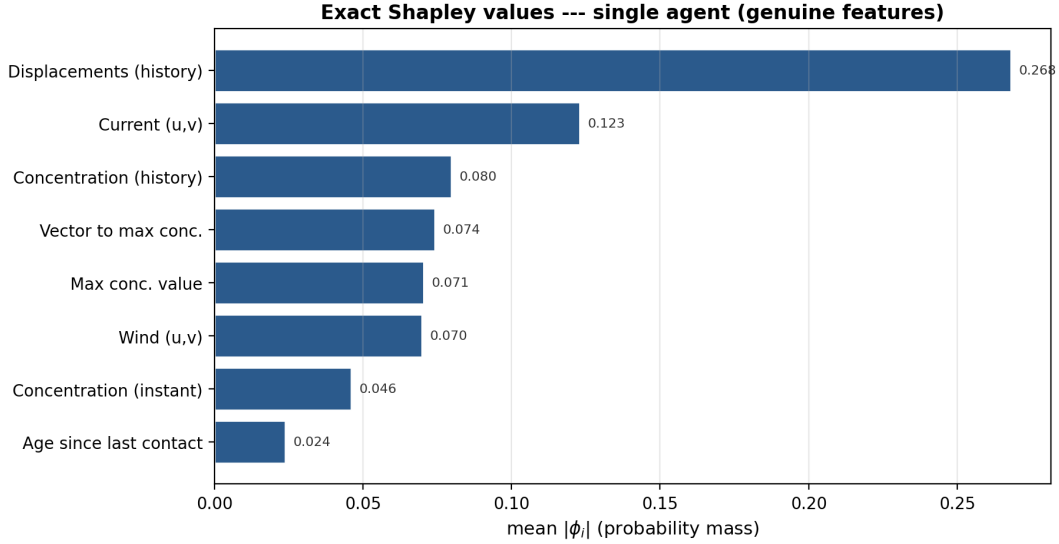


Figure 3: Exact Shapley values — single agent (genuine, non-masked groups). Each bar is a group’s mean absolute Shapley value $|\phi_i|$: its average marginal contribution to the chosen-action probability across all 2^{10} coalitions, over 1500 states. Larger = the policy attributes more to that group; the displacement memory is the dominant player.

5 Explainability of the model *with* formation

For the comparison we run the same full protocol on the model with formation (double ring, $v_{\max}=2$).

5.1 Causal group ablation

Ablated group	Δ SR	95% CI	Δ steps
Ring 20 m (history)	-26.8	$[-30.3, -23.4]$	+78%
Displacements (history)	-12.8	$[-15.7, -10.3]$	+29%
Ring 20 m (instant)	-6.2	$[-8.5, -4.5]$	+35%
Vector to max conc.	-5.0	$[-6.7, -3.4]$	+34%
Current (u, v)	-3.0	$[-4.5, -1.8]$	+49%
Wind (u, v)	-2.2	$[-3.5, -1.0]$	+20%
Concentration (history)	-0.5	$[-1.1, +0.2]$	+2%
Ring 50 m (history)	-0.3	$[-0.8, +0.0]$	+14%
Ring 50 m (instant) / Conc. (instant)	≈ 0	—	—
Max conc. value / Age	≈ 0	—	—

Table 5: Causal group ablation on the model with formation (baseline SR 99.8%).

With the formation active the picture is the opposite of the single agent: the most load-bearing group is the **20 m ring history** (-26.8 SR points), i.e. the network integrates over time the *spatial* chemical field read by the sensors. The displacement memory is second (-12.8) and the max-concentration vector third (-5.0). Three groups are indistinguishable from baseline (instantaneous concentration, its history, the 50 m ring instantaneous): the sensor ring makes them redundant.

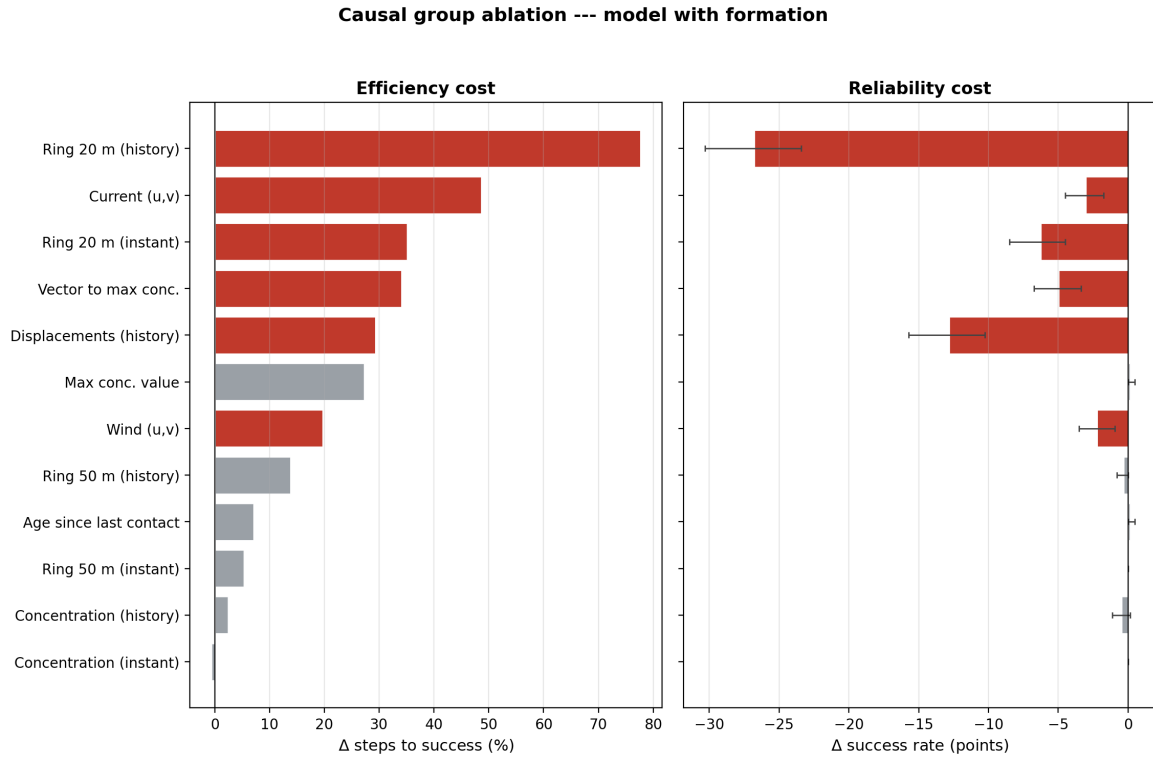


Figure 4: Group ablation — model with formation. *Left*: efficiency cost (extra steps to success, %); *right*: reliability cost (SR drop, points) with 95% bootstrap CIs; red when the interval excludes zero, grey when it contains it. The 20 m ring history is by far the dominant group; the 50 m ring history carries only a small efficiency effect.

5.2 Exact Shapley values

By Shapley value the ranking is led jointly by **current** and **displacements** ($|\phi| = 0.236$), followed by the 20 m ring history (0.183) and instant (0.104); the 50 m ring contributes little (0.058/0.031). Full observation raises the probability of the chosen action from 0.20 (baseline) to 0.71.

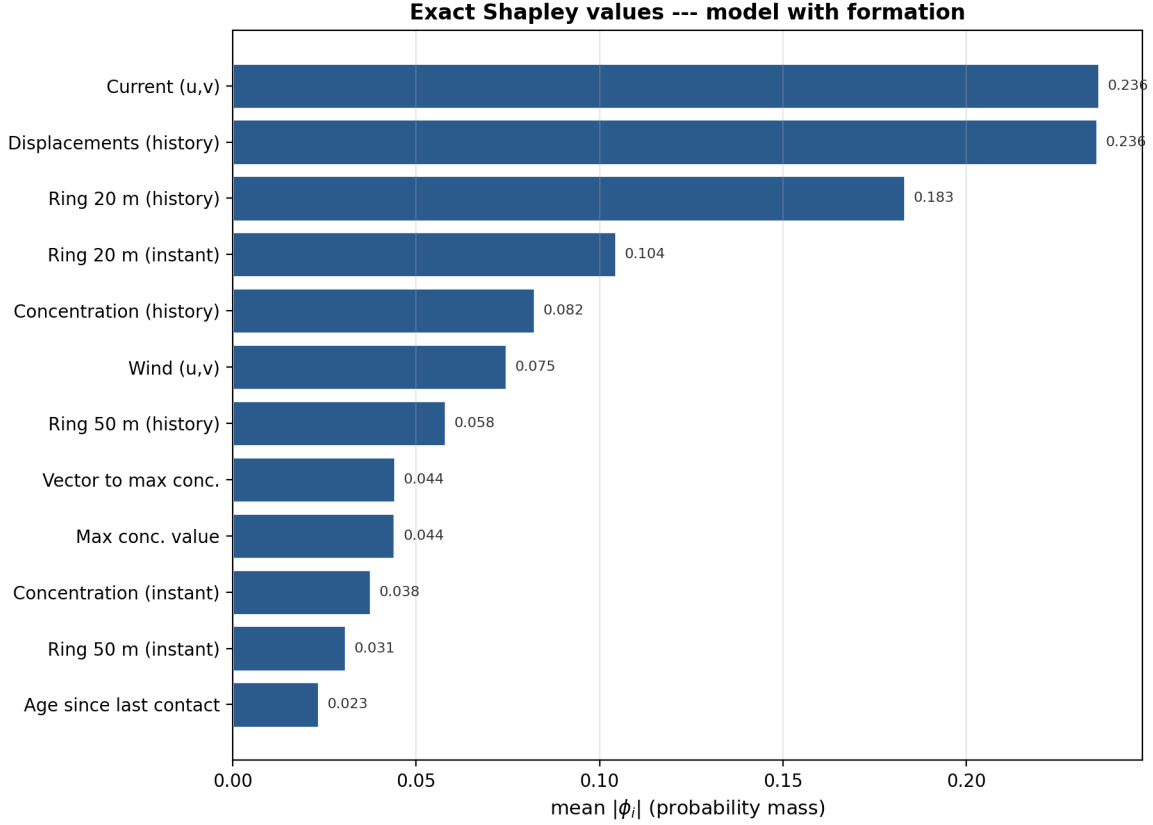


Figure 5: Exact Shapley values — model with formation (mean $|\phi_i|$ per group, same definition as Figure 3). Current and displacements lead, closely followed by the 20 m ring history; the 50 m ring contributes little.

Ablation vs. Shapley: why the two rankings differ. The two panels answer *different* questions, and their divergence is informative, not contradictory. Ablation (Figure 4) measures **causal necessity** — what breaks when a group is removed — whereas the Shapley value (Figure 5) measures **attribution** — how much the policy’s output depends on a group on average. The clearest case is the **current**: one of the highest Shapley values (0.236) yet almost no ablation effect (-3.0). The policy *watches* the current but does not *need* it: its effect is already encoded in the plume shape read by the ring, so removing it leaves the agent able to recover the same information elsewhere — it is *redundant*. The **ring history**, conversely, dominates the ablation (-26.8) because no other group carries the instantaneous spatial gradient: it is *irreplaceable*. Displacements score high on both — watched *and* needed. In short, “what the network looks at” is not “what the task needs”, and comparing the two is exactly what exposes the redundant channels (current, wind) and the load-bearing ones (the ring).

6 Conclusions

- **The formation has a measurable but contained value.** At parity of everything, removing it and retraining costs -6.4 **points** of global SR and $\sim 45\%$ in speed, concentrated in the advected/diffuse-plume scenarios (V2 -12.8 , Q1/2 -9.0), where the instantaneous spatial gradient is most valuable.

- **Learning matters more than the formation.** The single agent without formation (93.3%) beats by +35 **points** the FCM that has the formation (58.2%). What separates success from failure is not the sensor, but the learned strategy.
- **How the single agent compensates.** The explainability shows a clear shift from *spatial* to *temporal* sensing: deprived of the ring, the agent relies on the displacement memory (−55.6), on homing to the maximum concentration seen (−25.8) and on its own concentration history (−11.2) — reconstructing in time the gradient the formation provided in space. With the formation, the same policy leans on the ring history (−26.8), integrating the sensors over time rather than chasing the instantaneous brightest one (sensor→action lift only 1.06×).
- **Method.** Retraining ablation gives a fairer measure than frozen-model ablation (which overstates importance): the agent can adapt, and the residual drop is the *irreducible* value of the formation. (Caveat: masking a channel via `VecNormalize` turns it into a constant, which must be excluded from the ablation — see Section 4.)